

GridLine AI

A disaster intelligence brain for a city's emergency operations centre.

In a monsoon storm, the facts that decide whether a hillside fails are scattered: a rain gauge, a soil probe, a construction permit's halt conditions, a landslide report from three years ago, and last year's change to a drainage culvert. A duty officer needs hours to join them. By then the slope may already be moving.

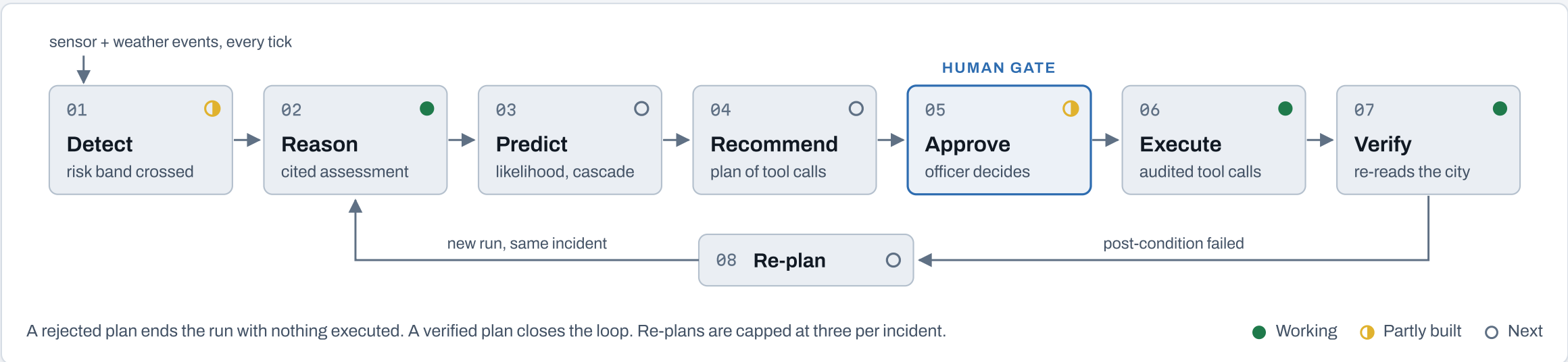
GridLine watches the city continuously. When a threat starts to form, it reasons over live readings and the city's own documents, cites a source for every claim, and proposes preventive actions. **An officer approves or rejects each one.** GridLine then carries out the approved actions through audited tools and reads the city back to prove they took effect. If they did not, it re-plans.

BUILT FOR	RUNS	STATUS
Emergency operations duty officers	Offline, with no API key, on a laptop	Core loop half built; smoke test green

HOW IT WORKS

One loop per threat, run as a LangGraph agent

Each incident gets its own graph run. Every step publishes its output to the dashboard as it happens, so the officer watches the reasoning rather than receiving a verdict.



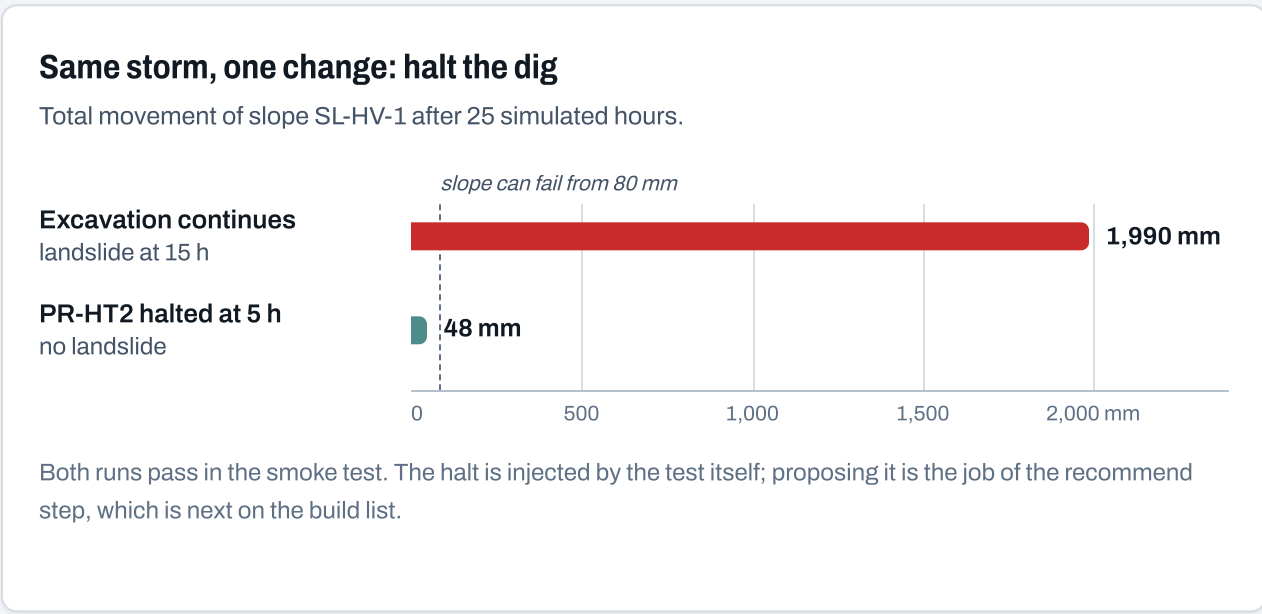
PRIMARY DEMO SCENARIO

A construction site keeps digging into a wet hillside

Hillview Terrace Phase 2 (project PR-HT2, permit HT-2026-014) is excavating slope SL-HV-1, a 32° hillside in the fictional city of Nandipur, as monsoon rain builds. If the slope fails, the debris blocks the D-7 drain and the water backs up into Riverside below. The table is the scenario as the simulator plays it, 5 simulated minutes per tick.

SIM TIME	WHAT HAPPENS	RAIN 24 h, mm	SLOPE saturation	SLOPE MOVED total, mm	D-7 LOAD flow ÷ capacity	RIVERSIDE water, cm	KALINADI river, m	POLICY LINE CROSSED
0–3 h	Excavation continues in light rain	9	0.50	0	0.12	0.0	1.68	None
3–8 h	Rain intensifies	46	0.50	0	0.40	0.0	1.87	None
8–12 h	Slope starts to creep	99	0.73	1	0.51	0.0	2.18	Construction must halt: rain ≥ 65 mm or saturation ≥ 0.70 dmp-2024#s5 permit-ht-2026-014#s3
12–15 h	Slope critical	144	0.92	45	0.52	0.0	3.12	Landslide critical: saturation ≥ 0.85 dmp-2024#s4.2
15–17 h	Landslide blocks D-7 and Hill Road	174	1.00	1,635	1.76	14.8	3.95	D-7 treated as overflowing: flow above capacity pol-flood-response-2024#s3
17–25 h	Riverside floods, Riverside Bypass cut	245	1.00	1,990	1.76	41.9	5.21	Roads close at 30 cm water; Kalinadi past warning stage pol-road-closure-2021#s3 dmp-2024#s4.3

Coloured values have reached a band in the city's Disaster Management Policy: Watch Warning Critical Stage peaks from scripts/demo_smoke.py , seed 42; slope movement at stage end.



What GridLine has to get right here

- Join the evidence.** Rain at gauge RG-02, saturation at SM-01, slope movement, the permit's halt conditions, the 2019 and 2022 Hillview landslide reports, and the 2025 Hill Road widening that cut D-7's capacity from 42 to 27 m³/s.
- See the cascade.** A slope failure here blocks D-7, which floods Riverside and cuts both Hill Road and the Riverside Bypass.
- Recommend in time.** Propose `create_construction_restriction(PR-HT2, halt)` with its citations, before the slope reaches the critical band.
- Prove it.** After approval, read project PR-HT2 back as `halted` and its excavation depth unchanged.

WHY AN OFFICER CAN TRUST IT

Four rules the code enforces

Every claim cites a source, or it is rejected

Outputs are structured. Each claim carries the IDs of the document sections or live readings it rests on, and a validator rejects any ID that was not in the reasoner's input.

e.g. [dmp-2024#s4.2](#) [rep-2022-1s-01#s3](#)

A person approves anything consequential

Nine tools always need a recorded decision, and public alerts need one at warning level and above. The executor refuses them without an approval ID, in demo mode too.

Inspections, monitoring tasks and incident records run automatically.

The AI changes the city only through tools

21 typed tools cover crews, ambulances, shelters, hospital beds, roads, construction, alerts, evacuation and incidents. Every call is idempotent and audited with before and after snapshots.

e.g. `close_road` `open_shelter` `reserve_hospital_beds`

Success is measured, not reported

After an action, the tool re-reads the database against its declared post-condition. Anything short of verified goes back to re-plan with the concrete failure, such as "crew C-4 target zone: expected Z-HV, observed Z-RS".

DATA

Two data modes, one pipeline

DEMO · DEFAULT Nandipur, a fictional city 10 zones with slopes, rivers, drains, roads, sensors, crews, shelters and hospitals. 37 documents: 8 policies, 4 SOPs, 17 past-disaster reports, a permit, surveys and an infrastructure change log. Four scripted scenarios, deterministic by seed. No network needed.	LIVE Kalyan-Dombivli, Maharashtra Real modelled weather from Open-Meteo for 19.235° N, 73.13° E, polled every 5 minutes with no API key. A failed fetch shows as unavailable and nothing is filled in. Weather only: no sensors, crews or map for the real city.
--	---

Both modes publish the same normalized events. Nothing after the event bus knows which one is running.

BUILD STATUS

What runs today, and what comes next

- WORKING

Simulation of rain, soil, drains and river; 4 scenarios

Event bus, event log and WebSocket stream

RAG over 37 documents with citation IDs

Knowledge graph over the city tables

Reasoning: observe, retrieve, assess, with citation checks

21 tools with approval rules, audit and verification

Dashboard and the DEMO / LIVE switch

Smoke test: 7 of 7 checks, 5,853 events
- NEXT

Threat detector: zone risk indices that open incidents

Predict, cascade, recommend graph steps

Approval gate in the graph, plus the decision API

Execute, verify, re-plan wired into the graph

Claude provider with structured outputs

Pump deployment tool for D-7

UNDER THE HOOD

One process, one database, one frontend

BACKEND	STORAGE	REASONING	FRONTEND
Python 3.13, FastAPI in a single process: simulation, event bus, agent runner and API together. No queues or extra services.	PostgreSQL 16 + pgvector holds the city, the append-only event log, the RAG index and the agent's checkpoints.	LangGraph agent, local embeddings (bge-small, CPU), and Claude through the Anthropic SDK. A deterministic reasoner that reads the same inputs keeps it working offline.	React 19, TypeScript, Tailwind, fed by REST and a live WebSocket. API types are generated from the backend schema.